CALCULUS I - AY2024-25

Chapter 6: Sequences and Series

Outline

1. Sequences, Convergence, Limits
2. Infinite Series
3. Properties of Series, Tests for Convergence
4. Power Series, Taylor & Maclaurin Series

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First we introduce some notations.

- $\mathbb{N} = \{1, 2, 3, \cdots\}$, the set of natural numbers (positive integers).

  We use $\mathbb{N}_0$ to denote the set $\mathbb{N} \cup \{0\} = \{0, 1, 2, 3, \cdots\}$.

- $\mathbb{Z} = \{0, \pm 1, \pm 2, \pm 3, \cdots\}$, the set of integers.

- $\mathbb{Q} = \left\{ \frac{p}{q} : p \in \mathbb{Z},\ q \in \mathbb{N},\ \gcd(p, q) = 1 \right\}$, the set of rational numbers.

• Sequences

Definition

A sequence is a set of real numbers written in a definite order

$$a_1, a_2, a_3, \ldots, a_n, \ldots$$

It is usually denoted by $\{a_n\}$, $\{a_n\}_{n=1}^{\infty}$, (a_n), or $(a_n)_{n=1}^{\infty}$. In general, a_n is called the nth *term* of the sequence.

The index of a sequence may start from some integer $n_0 \geq 0$:

$\{a_n\}_{n=n_0}^{\infty}$ or $a_{n_0}, a_{n_0+1}, a_{n_0+2}, \ldots$

Note. A sequence of real numbers can be considered as a function from say $\mathbb{N}$ to $\mathbb{R}$. So given a function $f : \mathbb{N} \to \mathbb{R}$, $f(n)$ is just the nth term in the sequence.

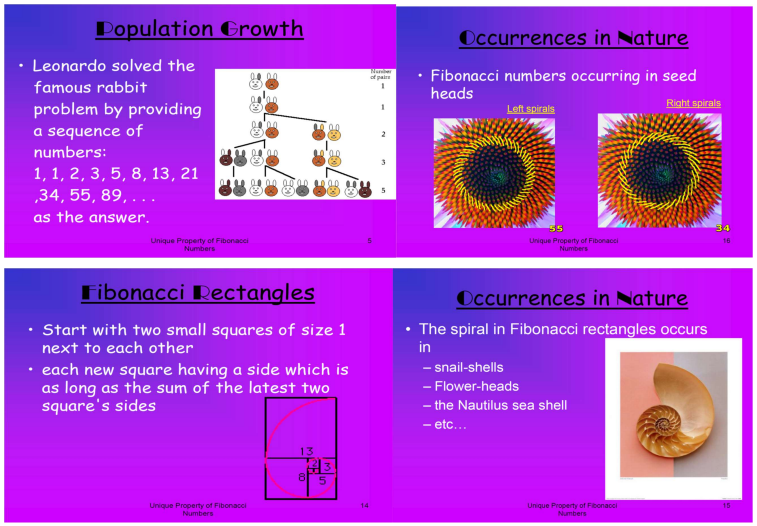
• Different ways to define sequences

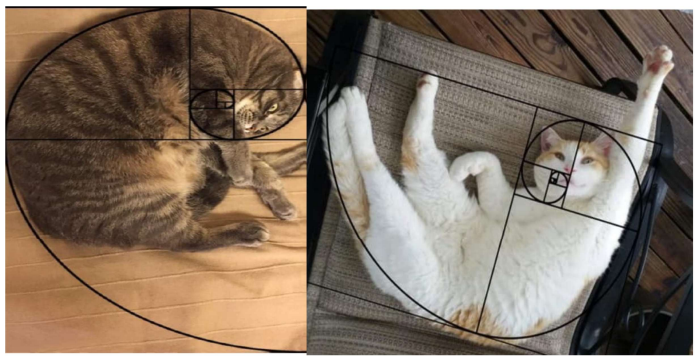
1. Sequences can be defined by giving a formula for the nth term.
2. Sequences may be defined by different expressions for different subsets of indices (like piecewise functions).
3. Sequences may be defined recursively.
4. Some sequences do not have a simple defining expressions.

Example

1. $\{(-1)^n\}_{n=1}^{\infty} = \{-1, 1, -1, 1, \ldots\}$: $a_n = (-1)^n$
2. $\{c_n\}$, with $c_n = \begin{cases} 1 & \text{if } 1 \leq n \leq 100, \\ \ln(\sqrt{n} - 10) & \text{if } n \geq 101. \end{cases}$
3. The Fibonacci sequence: $F_1 = F_2 = 1$, $F_n = F_{n-1} + F_{n-2}$ $(n \geq 3)$.
4. The sequence $\{p_n\}$, where p_n is the n-th prime number.

Web-search. Find out how Fibonacci sequence arises.





• Convergent sequences and limits

We consider the following examples.

(a) The sequence $\left\{\dfrac{n}{n+1}\right\}$ approaches the number 1 as n becomes larger. We say that this sequence converges to 1, and write $\lim\limits_{n\to\infty}\dfrac{n}{n+1}=1$.

(b) The sequence $\{(-1)^n\})$ does not approach any number as n becomes larger. We say that this sequence diverges.

(c) The sequence $\left\{\sqrt{n-3}\right\}_{n\geq 3}$ becomes larger and larger unboundedly as n becomes larger. It does not approach any number as n becomes larger, and we write $\lim\limits_{n\to\infty}\sqrt{n-3}=\infty$. We also say that this sequence diverges.

We have the following formal definition of limit of a sequence.

Definition

A sequence $\{a_n\}$ converges to a real number L, or $\{a_n\}$ has limit L, if

> for every number $\varepsilon > 0$, there exists a corresponding integer N such that for all $n > N$,
>
> $$|a_n - L| < \varepsilon.$$

In this case, we write $\lim_{n\to\infty} a_n = L$ or $a_n \to L$ as $n \to \infty$, and say that the sequence converges to L or is convergent with limit L.

Note. The value of N usually depends on the value of ε, i.e. $N = N(\varepsilon)$. The smaller ε is, the larger N may have to be.

If such a number L does not exist, we say that the sequence diverges or is divergent.

(The limit of a sequence can be illustrated as a "cricket-trap" jumping.)

Example

Suppose $0 < |r| < 1$. Prove that

$$\lim_{n\to\infty} r^n = 0.$$

Solution. Let $\varepsilon > 0$ be an arbitrarily given positive number.

Our aim is to find an integer N so that for all $n > N$: $|r^n - 0| = |r|^n < \varepsilon$.

This requires for all $n > N$: $n \ln |r| < \ln \varepsilon \iff n > \dfrac{\ln \varepsilon}{\ln |r|}$ (since $|r| < 1$).

Hence, we may take $N \in \mathbb{N}$ satisfying $N > \dfrac{\ln \varepsilon}{\ln |r|}$ to get the result.

• A typical procedure to prove $\{a_n\} \to L$

Summary

1. Let $\varepsilon > 0$ be any number.
2. Choose an N big enough.

   We need to find N by several-step calculations.

   (This is the main part of the proof)

3. Verify that N works, i.e.

   For all $n > N$, show that $|a_n - L| < \varepsilon$.

4. Conclude that $\{a_n\} \to L$ as $n \to \infty$, i.e. $\lim_{n\to\infty} a_n = L$.

Using the definition of limit, we are able to prove the uniqueness of limit, and other properties of limits.

- **Uniqueness of the limit**

Theorem

If a sequence $\{a_n\}$ converges, then its limit is unique. In other words, if $\lim_{n\to\infty} a_n = L$ and $\lim_{n\to\infty} a_n = L^*$, then $L = L^*$.

- **Properties of limits**

Theorem

Suppose that $\lim_{n\to\infty} a_n = a$, $\lim_{n\to\infty} b_n = b$. Then sequences $\{Ca_n\}$ $(C \in \mathbb{R})$, $\{a_n \pm b_n\}$, $\{a_n b_n\}$, and $\left\{\frac{a_n}{b_n}\right\}$ (provided $b \neq 0$, $b_n \neq 0$ for all n), are convergent.

Moreover, the limits are respectively Ca, $a \pm b$, ab, and $\frac{a}{b}$.

Example

We know that $\lim_{n\to\infty} \frac{1}{n} = 0$ and $\lim_{n\to\infty} \frac{n}{n+1} = 1$.

Using the properties, we have

1. $\lim_{n\to\infty} \left(\frac{1}{n} + \frac{n}{n+1}\right) = \lim_{n\to\infty} \frac{1}{n} + \lim_{n\to\infty} \frac{n}{n+1} = 0 + 1 = 1$
2. $\lim_{n\to\infty} \frac{1}{n^k} = \lim_{n\to\infty} \left(\frac{1}{n}\right)^k = \underbrace{\left(\lim_{n\to\infty} \frac{1}{n}\right) \cdots \left(\lim_{n\to\infty} \frac{1}{n}\right)}_{k \text{ times}} = 0$, for $k \in \mathbb{N}$.

- **Interchange order**

Theorem

Suppose f is a continuous function. Then

$$\lim_{n\to\infty} f(a_n) = f\left(\lim_{n\to\infty} a_n\right),$$

provided that the limit in the left-hand side exists and the function f is defined at this limit value.

Example

1. $\lim\limits_{n\to\infty} \sin\dfrac{1}{n} = \sin\left(\lim\limits_{n\to\infty} \dfrac{1}{n}\right) = \sin 0 = 0$
2. $\lim\limits_{n\to\infty} \sqrt{\dfrac{3n}{n+1}} = \sqrt{\lim\limits_{n\to\infty} \dfrac{3n}{n+1}} = \sqrt{3}$

- **Divergent sequences**

A sequence $\{a_n\}$ is said to be **divergent** if it does not converge.

- **Subsequences**

Definition

A **subsequence** of $\{a_n\}$ is a sequence "extracted" from the sequence $\{a_n\}$.

For examples,

- $\{a_{n^2}\}$, that is $a_1, a_4, a_9, a_{16}, \ldots$
- The even subsequence $\{a_{2n}\}$, that is $a_2, a_4, a_6, a_8, \ldots, a_{2n}, \ldots$

  The odd subsequence $\{a_{2n-1}\}$, that is $a_1, a_3, a_5, a_7, \ldots, a_{2n-1}, \ldots$

- **Useful property of subsequences**

Theorem

If $\lim_{n\to\infty} a_n = L$, *then every subsequence of* $\{a_n\}$ *also converges to* L.

- **How to prove a divergence?**

As a consequence, we may conclude that $\lim_{n\to\infty} a_n$ does not exist in each of the following cases:

(a) There is a divergent subsequence of $\{a_n\}$.

(b) There are two convergent subsequences of $\{a_n\}$ which converge to different limits.

Example

(a) For the sequence $a_n = \begin{cases} \ln \frac{1}{n} & \text{if } n = 3k, \\ \pi & \text{otherwise,} \end{cases}$

we have the subsequence $\{a_{3n}\} = \left\{\ln \frac{1}{3n}\right\}$ diverges.

So the given sequence $\{a_n\}$ diverges.

(b) For the sequence $\{(-1)^n\}$, i.e. $a_n = (-1)^n$, we have

– the odd subsequence $\{a_{2n-1}\} = \{-1\} \implies \lim_{n\to\infty} a_{2n-1} = -1$,

– the even subsequence $\{a_{2n}\} = \{1\} \implies \lim_{n\to\infty} a_{2n} = 1$.

So the given sequence $\{a_n\}$ diverges.